

GRAVITY — Full System Documentation

Trackalways Limited · Family Safety & Location Tracking Platform Last updated: June 2026

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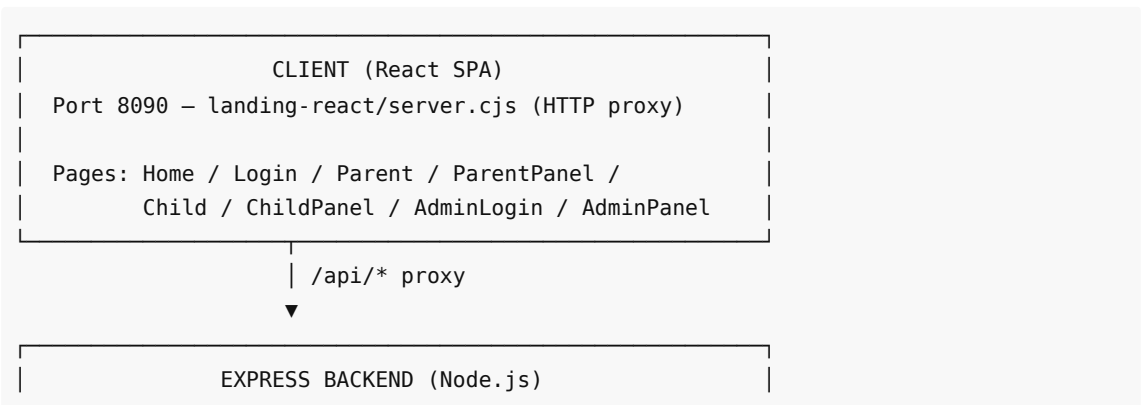
1. SYSTEM OVERVIEW

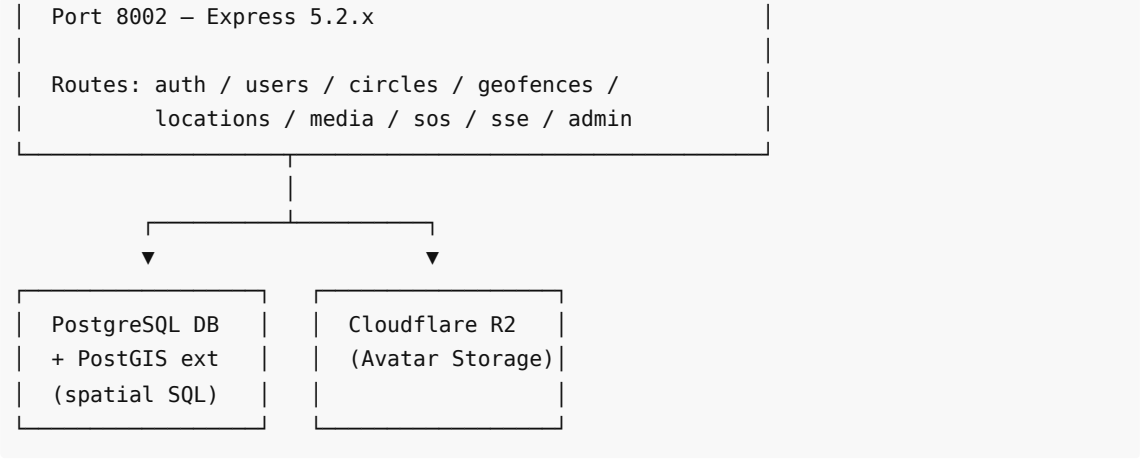
Project Name: Gravity (brand) / Trackalways (company name) **Type:** Web-based family safety & location tracking platform **Target:** Mobile-first responsive web app (phone-frame UI on desktop, fullscreen on mobile) **UI Theme:** Dark green premium (#050C08 background, #00E676 accent) **Font:** Plus Jakarta Sans **Company:** Trackalways Limited **Copyright:** © 2026 Trackalways Limited

Core Concept

Parents can monitor their children and family members in real-time on a map. Children can send SOS alerts in emergencies. Family circles group members together. Safe zones (geofences) trigger automatic alerts on entry/exit.

2. ARCHITECTURE





Tech Stack

Layer	Technology
Frontend	React 18 + TypeScript + Vite
Routing	React Router v6
Maps	Leaflet.js (loaded via CDN)
Backend	Express.js 5.2.x (Node.js)
Database	PostgreSQL + PostGIS extension
Auth	JWT (jsonwebtoken) + bcryptjs
Validation	Zod (backend schema validation)
SMS	MSG91 (OTP delivery)
Storage	Cloudflare R2 (S3-compatible)
Real-time	Server-Sent Events (SSE)
Styling	CSS Modules (per-component)

3. ROUTING & PAGES

/	→ Home.tsx	(Landing page)
/login	→ Login.tsx	(Phone OTP login/register)
/parent	→ Parent.tsx	(Parent marketing page)
/parent/panel	→ ParentPanel.tsx	(Parent dashboard – auth required)
/parent-panel	→ redirect → /parent/panel	
/child	→ Child.tsx	(Child marketing page)
/child/panel	→ ChildPanel.tsx	(Child dashboard – auth required)
/child-panel	→ redirect → /child/panel	
/admin	→ redirect → /admin/login	
/admin/login	→ AdminLogin.tsx	(Admin password login)
/admin/panel	→ AdminPanel.tsx	(Admin dashboard – admin auth required)

Server (landing-react/server.cjs)

- Runs on port **8090**
- Proxies all `/api/*` requests to backend at port **8002**
- Handles SSE streaming correctly (no buffering, keep-alive)
- SPA fallback: serves `index.html` for all unknown routes

4. AUTHENTICATION SYSTEM

User Authentication (Phone OTP)

1. **Send OTP:** `POST /api/v1/auth/send-otp` → generates 6-digit OTP, stores in `phone_otp`, sends via MSG91
2. **Rate limit:** Max 3 OTPs per phone per 10 minutes
3. **OTP expiry:** 10 minutes
4. **Login:** `POST /api/v1/auth/login` → verifies OTP + password → returns JWT
5. **Register:** `POST /api/v1/auth/register` → verifies OTP → creates user → returns JWT

JWT Storage: `localStorage.gravity_token` **User Object:** `localStorage.gravity_user` (JSON: id, name, phone, email, role, avatar_url, created_at) **Auth Header:** `Authorization: Bearer <token>`

Google Sign-In

- Google GSI (Google Sign-In) script loaded dynamically
- `window.handleGoogleCredential` callback processes Google JWT
- Falls back to manual Google sign-in button if one-tap doesn't appear in 2.5s

Admin Authentication

1. **Login:** `POST /api/v1/admin/login` with `{ password }` → checked against `ADMIN_SECRET` env var
2. **Returns:** JWT with `{ role: 'admin' }` payload, expires in 12 hours
3. **Storage:** `localStorage.admin_token`
4. **Auth Header:** `x-admin-token: <token>` (all admin API calls)
5. **Token check on load:** `AdminLogin.tsx` auto-redirects to `/admin/panel` if valid token exists

Auth Guards

- `ChildPanel`: if no `gravity_token` → redirect to `/login?redirect=/child/panel`
- `ParentPanel`: if no `gravity_token` → redirect to `/login?redirect=/parent/panel`
- `AdminPanel`: if no `admin_token` → redirect to `/admin/login`
- 401 responses → clear `localStorage` → redirect to login

5. CHILD PANEL

URL: `/child/panel` **Auth:** JWT required (`gravity_token`) **UI:** Phone-frame mockup (max-width 430px), 6-tab bottom navigation

Bottom Navigation Tabs

 Home |  Map |  SOS (center, floating red) |  Family |  Alerts |  Profile

TAB 1: HOME

Purpose: Quick status overview + emergency SOS

Sections:

1. **Status Bar** — clock (12-hr), signal bars, WiFi, battery (CSS animated)
2. **Header** — Gravity logo (SVG pin), "Welcome back / Hey, {Name}" + avatar
3. **My Status Card** — greeting (morning/afternoon/evening + emoji), "Your family can see your location" with pulsing green dot
4. **Location Card** — "Location sharing active" + last updated time + live lat/lng coords + LIVE badge
5. **SOS Button (Home)** — 140px red circle, 3 animated rings, hold 3 seconds to activate
 - Countdown: 3 → 2 → 1 with flash animation, releases = cancels
 - Sends: GPS coordinates + "SOS! I need help!" to all circle members
6. **Quick Actions** (2×2 grid):
 -  **I'm Safe** → `POST /api/v1/sos/safe` + toast
 -  **Share Location** → copies `{origin}/share?uid={id}` to clipboard
 -  **Arrive Safe** → shows coordinates in toast (notification placeholder)
 -  **Message Family** → switches to Family tab
7. **Family Circle strip** — horizontal scroll, avatar rings with online/offline dot
 - Own avatar (always first, green glow)
 - Other members with color-coded rings
 - Click → shows name/status/coordinates in toast

Logic:

- `updateTime()` runs every 30s, updates greeting + status bar clock
 - GPS watch via `navigator.geolocation.watchPosition`, throttled to POST every 30s
 - Battery via `navigator.getBattery()` API, fallback 75%
-

TAB 2: MAP

Purpose: See all family members on live map

Features:

- Leaflet.js map (dark CDN tiles by default)
- **4 tile types:** 🌙 Dark / ☀️ Light / 🗺️ Satellite / 🏠 Street
- **Member markers:** Avatar photo circles with color-coded rings + name labels
 - Own marker: 52px, labeled "📍 You"
 - Others: 44px, labeled with first name
 - Popup on click: name, role, battery%, status
- **Zoom controls:** + / - buttons (custom, not Leaflet default)
- **Auto-fit:** `fitBounds` to show all located members
- **Nearest Member Card** — shows first other member's avatar + location status
- **Distance badges** — colored dot + name for each member

Map Init Logic:

- Map only initializes on first tab switch to 'map'
 - Leaflet CSS injected dynamically if not present
 - SSE `location_update` events move markers in real-time without reload
-

TAB 3: SOS (Center Tab)

Purpose: Emergency alert system

Sections:

1. **GPS Status Bar** —  Active (green) /  Off (red)
 2. **Big SOS Button** — 160px, 3 animated rings, "PRESS & HOLD" instruction
 - Same hold-3-seconds logic as Home SOS
 - GPS coordinates fetched fresh on activation
 3. **Quick Messages** (2×2 grid):
 -  "I need help!" (red)
 -  "I'm in danger!" (orange)
 -  "Call me now!" (blue)
 -  "Come pick me up!" (green)
 - Each: `POST /api/v1/sos` with message + current GPS
 4. **Emergency Contacts** — all other circle members
 - Avatar, name, role, online status
 - **Call button** → `tel:` link (if phone stored) or toast warning
 - **Emergency Services** card → always shows "112 · Police / Ambulance" → `tel:112`
-

TAB 4: FAMILY

Purpose: View all circle members

States:

- `hasCircle === null` → "Loading..."
- `hasCircle === false` → Join Circle box:
 -  QR code placeholder
 - Invite code input (A-Z 0-9 only, 12 chars max)
 - `POST /api/v1/circles/join` → reloads page on success
- `hasCircle === true` → Member cards

Member Card:

- Avatar with online/offline ring
 - Name (with "(You)" label for self)
 - Role (parent/child/member/admin)
 - Battery bar + percentage (green >60%, amber >20%, red ≤20%)
 - Online status dot
 - "  Located" badge (top-right, if coordinates available)
 -  **View on Map** button → switches to Map tab + centers on member
-

TAB 5: ALERTS

Purpose: Notification history for geofence & SOS events

Header:

- "Alerts" title + filtered notification count
- "Clear All" button (appears when alerts exist)

Filter Tabs:

- **All** — all alerts combined
- 📍 **Geofence** — only geofence entry/exit events
- SOS **SOS** — only SOS alerts

Empty States (filter-aware):

- All/Geofence: 📍 icon
- SOS: SOS icon
- Context-appropriate subtitle text

Alert Card:

- Left border color: red (SOS), green (geofence entry), amber (geofence exit)
- Icon: SOS or 📍
- Username + timestamp (relative: "Just now", "5m ago", "2h ago", "15 Jun")
- Message text

Real-time: SSE `geofence_event` and `sos_alert` events push new alerts **Unread badge:** Red dot on Alerts nav tab, count resets when tab opened

TAB 6: PROFILE

Purpose: Account management + settings

Sections:

My Account Card (dark card):

- Clickable avatar → file picker → R2 upload flow
 - Camera overlay on hover
 - Upload spinner during upload
 - ➡ edit badge (bottom-right)
- MEMBER/PARENT/CHILD role badge (green)
- Editable name input with "DISPLAY NAME" label
- Info rows (icon + subtle bg): ✉ Email / 📞 Phone / 📅 ¹⁷ Member since
- ✓ Save Profile button (full-width, gradient green)
 - PATCH `/api/v1/users/me` → updates name
 - Falls back to localStorage update if 404
- 🚪 Logout button (full-width, red outline, outside card)

Location Sharing (toggles):

- 📍 Share my location (on/off)
- 🎯 Location Precision: Exact / Approximate (segmented control)
- 📶 Auto-share in SOS zone (on/off)
- All toggles persist to `localStorage.gravity_toggles`

Notifications (toggles):

- 🏠 Family arrivals
- SOS SOS alerts from family
- 📍 Geofence alerts

Today's Activity (3 stat cards):

- Distance Today → `GET /api/v1/users/me/stats` → `today.distance_km`
- Safe Zones Visited → `today.safe_zones_visited`
- Family Check-ins → `today.family_checkins`
- Shows "—" if no data

Location History:

- Refresh button → `GET /api/v1/users/me/location-history` → last 10 entries
- Entry: 📍 icon + distance from home OR coordinates + date/time + 🔋 battery%
- Auto-loads when Profile tab opened

Bottom:

- 🏠 Back to Home link
 - "Gravity v1.0.0" version badge
-

6. PARENT PANEL

URL: `/parent/panel` **Auth:** JWT required (gravity_token) **UI:** Full-page web app with left sidebar navigation (not phone-frame)

Sidebar Navigation

- 🏠 Dashboard
 - 🗺️ Map (Live Map)
 - 👥 Members
 - 📌 Geofences (Safe Zones)
 - 🔔 Alerts
 - ⚙️ Settings / Profile
 - 🚪 Logout (bottom)
-

SECTION 1: DASHBOARD

Stats Overview (4 stat cards):

- Total Members in circle
- Active Now (online members)
- Safe Zones count
- SOS Alerts (today)

Live Member Status Strip:

- Horizontal scroll of member avatar cards
- Each: avatar, name, battery, online/offline indicator

Recent Alerts Panel:

- Last few geofence/SOS events with timestamps
-

SECTION 2: LIVE MAP

- Leaflet.js map (same tile types as ChildPanel)
- All circle members plotted with color-coded avatar markers
- Real-time updates via SSE `location_update`
- Tile switcher: Dark / Light / Satellite / Street

- Zoom controls
 - "ALL SAFE" badge (top right)
 - Member distance badges (bottom strip)
 - Nearest member card
-

SECTION 3: MEMBERS

Member management for circle owner

Member Cards:

- Avatar, name, role, online/offline status
- Battery percentage + colored bar
- Last location coordinates (if available)
-  View on Map button
- × Remove member button (with confirmation dialog)
 - `DELETE /api/v1/circles/{circleId}/members/{memberId}`

Invite New Member:

- Circle invite code displayed
 - Copy button
 - QR code generation (visual)
 - Regenerate code button → `PATCH /api/v1/circles/{id}/invite`
-

SECTION 4: GEOFENCES (Safe Zones)

Create Safe Zones:

- Interactive Leaflet map with circle drawing
- Click map to place center point
- Adjust radius (slider/input)
- Name the zone
- `POST /api/v1/geofences` → creates zone
- Zone saved with: name, center_lat, center_lng, radius, circle_id

Zone List:

- Each zone: name, address, radius, active/inactive toggle
- Delete zone button
- Click zone → centers map on it

Zone Events (Recent):

- Entry/exit events for each zone with member name + timestamp
-

SECTION 5: ALERTS

Same alert list as ChildPanel but from parent perspective:

- Filter: All / Geofence / SOS
 - Each card shows which family member triggered it
 - SOS cards have higher visual priority (red border)
 - Dismiss individual alerts
-

SECTION 6: SETTINGS / PROFILE

- Edit profile name, avatar upload (R2)
- Circle management: rename circle, view invite code
- Notification preferences toggles
- Account section: email, phone, member since
- Logout

7. ADMIN PANEL

URL: /admin/panel **Auth:** Admin token required (x-admin-token header) **UI:** Full desktop web app — dark sidebar + content area

Sidebar Sections

-  Dashboard
-  Users
-  Circles
-  SOS Events
-  Geofence Logs
-  OTP Logs
-  System
-  Broadcast

SECTION 1: DASHBOARD

GET /api/v1/admin/dashboard

Stat Cards (real DB queries):

Stat	Source
Total Users	COUNT(*) FROM users
Parents	COUNT WHERE account_type='parent'
Children	COUNT WHERE account_type='child'
Banned	COUNT WHERE is_banned=TRUE
Total Circles	COUNT(*) FROM circles
Active Now	DISTINCT user_id WHERE updated_at > NOW()-5min
SOS Today	COUNT FROM sos_events WHERE created_at > NOW()-24h
Geofence Events	COUNT(*) FROM geofence_events
Location Points	COUNT(*) FROM device_locations

SECTION 2: USERS

GET /api/v1/admin/users?page=1&search=&limit=20

Features:

- Paginated user list (20 per page)
 - Search by name / phone / email (ILIKE)
 - **Table columns:** Name, Phone, Email, Role, Circles, Joined, Status, Actions
 - **Ban/Unban toggle** → PATCH /api/v1/admin/users/:id/ban (toggles is_banned)
 - **Delete user** → DELETE /api/v1/admin/users/:id (with confirmation dialog)
 - Banned users shown with red badge
 - Circle count shown per user
-

SECTION 3: CIRCLES

GET /api/v1/admin/circles

Table columns: Circle Name, Owner, Phone, Members, Safe Zones, Invite Code, Created **Actions:**

-  Regenerate invite code → PATCH /api/v1/admin/circles/:id/invite
 -  Delete circle → DELETE /api/v1/admin/circles/:id (with confirmation)
-

SECTION 4: SOS EVENTS

GET /api/v1/admin/sos

Table columns: User, Phone, Circle, Location (lat/lng), Message, Time, Status, Action **Actions:**

-  Resolve SOS → PATCH /api/v1/admin/sos/:id/resolve → marks resolved=TRUE
 - Unresolved events highlighted in red
 - Last 100 SOS events shown
-

SECTION 5: GEOFENCE LOGS

GET /api/v1/admin/geofences

Table columns: User, Phone, Zone Name, Circle, Event Type (Entry/Exit), Time

- Last 200 events
 - Entry = green chip, Exit = red chip
 - Sortable by time (DESC)
-

SECTION 6: OTP LOGS

GET /api/v1/admin/otps

Table columns: Phone, OTP Code, Expires At, Used (✓/✗), Created At

- Last 100 OTP records
 - Used for debugging SMS delivery issues
 - Shows raw OTP codes (dev/debug tool)
-

SECTION 7: SYSTEM INFO

GET /api/v1/admin/system

Displays:

- Database size (`pg_size_pretty`)
- All table names + live row counts
- Rate limit config (900s window, 1000 req max)
- Node.js version
- Server uptime (seconds → formatted)
- Connected SSE clients (live count)

Danger Zone:

- 🗑️ Purge Location History → `DELETE /api/v1/admin/locations/purge?days=30`
 - Input: days (default 30, min 1)
 - Deletes all `device_locations` older than N days
 - With double confirmation dialog

SECTION 8: BROADCAST

`POST /api/v1/admin/broadcast`

Features:

- Message input (text)
- Type selector: info / warning / alert
- **Send** → pushes SSE event `admin_broadcast` to ALL connected clients
- Shows recipient count (connected SSE clients)
- ChildPanel/ParentPanel can listen for `admin_broadcast` events

8. LANDING PAGE

URL: / **File:** `Home.tsx` (1424 lines)

Sections (top to bottom):

1. **NAV** — Logo, links (Features/How it Works/Pricing/Download), Login/Get Started CTA
2. **HERO** — Headline ("Keep Your Family Safe, Always"), family member strip, live Leaflet map (right column), map type switcher, animated member markers with connection lines, sidebar distance badges
3. **PAIN POINT** — "Are you worried about..." emotional messaging
4. **FEATURES** (6 cards with inline SVG illustrations):
 - 🗺️ Live Location Tracking
 - 🏠 Safe Zones (Geofence)
 - 👨‍👩‍👧 Family Circles
 - 🆘 Emergency SOS
 - 🔋 Battery Monitoring
 - 🔒 Privacy Controls
5. **APP SCREENSHOTS** — 5 phone mockups (Live Map, SOS, Geofence, Battery, Alerts)
6. **STATS BAR** — 50K+ families · 5 countries · 99.9% uptime · <2s updates
7. **HOW IT WORKS** — 4-step timeline (Create → Invite → Share → Stay Safe)
8. **FOR PARENTS / FOR CHILDREN** — Split feature comparison
9. **COUNTRIES** — Supported regions
10. **TESTIMONIALS** — User reviews
11. **PRICING** — Free Forever / Pro (₹99/month)
12. **DOWNLOAD** — Google Play + App Store badges

Landing Page Effects:

- Canvas particle network (animated dots + connection lines)
- Scroll reveal animations (IntersectionObserver)
- Stats counter animation on scroll (count up)
- Leaflet map with demo member markers + connection SVG lines
- Mobile hamburger menu

9. BACKEND API ROUTES

Base URL: `/api/v1`

auth.js — `/api/v1/auth/`

Method	Path	Auth	Description
POST	<code>/send-otp</code>	None	Send 6-digit OTP to phone (MSG91 or console log)
POST	<code>/login</code>	None	Verify OTP + password → JWT token
POST	<code>/register</code>	None	Verify OTP → create user → JWT token

users.js — `/api/v1/users/`

Method	Path	Auth	Description
GET	<code>/me</code>	JWT	Get current user profile
PATCH	<code>/me</code>	JWT	Update name/email/push_token
GET	<code>/search?phone=</code>	JWT	Find user by phone number
POST	<code>/location</code>	JWT	Save GPS point (lat/lng/accuracy/battery)
PATCH	<code>/location</code>	JWT	Update battery level only
GET	<code>/me/stats</code>	JWT	Today's activity (distance/safe zones/checkins)
GET	<code>/me/location-history</code>	JWT	Last 50 location points

circles.js — `/api/v1/circles/`

Method	Path	Auth	Description
GET	<code>/</code>	JWT	List my circles
POST	<code>/</code>	JWT	Create new circle
POST	<code>/join</code>	JWT	Join circle via invite code
GET	<code>/:id/members</code>	JWT	List circle members with locations
DELETE	<code>/:id/members/:userId</code>	JWT	Remove member from circle

PATCH	/:id/invite	JWT	Regenerate invite code
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geofences.js — /api/v1/geofences/

Method	Path	Auth	Description
GET	/	JWT	List geofences for my circles
POST	/	JWT	Create safe zone
DELETE	/:id	JWT	Delete safe zone
GET	/events	JWT	Recent geofence events

locations.js — /api/v1/locations/

Method	Path	Auth	Description
POST	/	JWT	Save location point (from Expo mobile)
GET	/:userId/history	JWT	Location history for user

sos.js — /api/v1/sos/

Method	Path	Auth	Description
POST	/	JWT	Trigger SOS — broadcasts to all circle members via SSE + logs to DB
POST	/safe	JWT	Send "I'm safe" notification
GET	/history	JWT	Last 50 SOS events

sse.js — /api/v1/sse/

Method	Path	Auth	Description
GET	/stream?token=	JWT (query param)	SSE connection — real-time event stream

media.js — /api/v1/media/

Method	Path	Auth	Description
POST	/avatar/presign	JWT	Get R2 presigned upload URL
POST	/avatar/confirm	JWT	Confirm upload + save publicUrl to user

admin.js — /api/v1/admin/

Method	Path	Auth	Description
POST	/login	None	Admin password login → JWT
GET	/dashboard	Admin	System-wide stats

GET	/users	Admin	Paginated user list with search
PATCH	/users/:id/ban	Admin	Toggle user ban status
DELETE	/users/:id	Admin	Delete user permanently
GET	/circles	Admin	All circles with owner info
DELETE	/circles/:id	Admin	Delete circle
PATCH	/circles/:id/invite	Admin	Regenerate invite code
GET	/sos	Admin	Last 100 SOS events
PATCH	/sos/:id/resolve	Admin	Mark SOS as resolved
GET	/geofences	Admin	Last 200 geofence events
GET	/otps	Admin	Last 100 OTP records
GET	/system	Admin	DB stats + server info + SSE count
DELETE	/locations/purge?days=	Admin	Purge old location data
POST	/broadcast	Admin	Broadcast SSE message to all clients

10. DATABASE SCHEMA

users

```

id          UUID PRIMARY KEY DEFAULT gen_random_uuid()
name        TEXT NOT NULL
phone       TEXT UNIQUE
email       TEXT UNIQUE
password_hash TEXT
account_type TEXT DEFAULT 'child' -- 'parent' | 'child' | 'admin'
country_code TEXT DEFAULT '+91'
avatar_url  TEXT
push_token  TEXT
is_banned   BOOLEAN DEFAULT FALSE
created_at  TIMESTAMPTZ DEFAULT NOW()
updated_at  TIMESTAMPTZ DEFAULT NOW()

```

circles

```

id          UUID PRIMARY KEY DEFAULT gen_random_uuid()
name        TEXT NOT NULL
invite_code TEXT UNIQUE NOT NULL
icon_url    TEXT
created_by  UUID REFERENCES users(id)
created_at  TIMESTAMPTZ DEFAULT NOW()

```

circle_members

```
circle_id  UUID REFERENCES circles(id) ON DELETE CASCADE
user_id    UUID REFERENCES users(id) ON DELETE CASCADE
role       TEXT DEFAULT 'member' -- 'admin' | 'member'
joined_at  TIMESTAMPTZ DEFAULT NOW()
PRIMARY KEY (circle_id, user_id)
```

device_locations (historical — PostGIS)

```
id          UUID PRIMARY KEY DEFAULT gen_random_uuid()
user_id     UUID REFERENCES users(id) ON DELETE CASCADE
geom        GEOMETRY(Point, 4326) -- PostGIS spatial point
accuracy    FLOAT
speed       FLOAT
bearing     FLOAT
altitude    FLOAT
battery_level INTEGER
recorded_at TIMESTAMPTZ DEFAULT NOW()
```

user_latest_locations (live snapshot — PostGIS)

```
user_id     UUID PRIMARY KEY REFERENCES users(id) ON DELETE CASCADE
geom        GEOMETRY(Point, 4326)
accuracy    FLOAT
battery_level INTEGER
updated_at  TIMESTAMPTZ DEFAULT NOW()
```

safe_zones (geofences)

```
id          UUID PRIMARY KEY DEFAULT gen_random_uuid()
circle_id   UUID REFERENCES circles(id) ON DELETE CASCADE
name        TEXT NOT NULL
address     TEXT
center_lat  FLOAT NOT NULL
center_lng  FLOAT NOT NULL
radius      INTEGER DEFAULT 100 -- meters
active      BOOLEAN DEFAULT TRUE
created_at  TIMESTAMPTZ DEFAULT NOW()
```

geofence_events

```
id          UUID PRIMARY KEY DEFAULT gen_random_uuid()
user_id     UUID REFERENCES users(id) ON DELETE CASCADE
safe_zone_id UUID REFERENCES safe_zones(id) ON DELETE CASCADE
```

```

event_type TEXT NOT NULL -- 'entry' | 'exit'
created_at TIMESTAMPTZ DEFAULT NOW()

```

sos_events

```

id          UUID PRIMARY KEY DEFAULT gen_random_uuid()
user_id     UUID REFERENCES users(id) ON DELETE CASCADE
user_name   TEXT
circle_id   UUID
latitude    FLOAT
longitude    FLOAT
message     TEXT
resolved    BOOLEAN DEFAULT FALSE
created_at  TIMESTAMPTZ DEFAULT NOW()

```

phone_otps

```

phone       TEXT NOT NULL
code        TEXT NOT NULL
expires_at  TIMESTAMPTZ NOT NULL
used        BOOLEAN DEFAULT FALSE
created_at  TIMESTAMPTZ DEFAULT NOW()

```

11. REAL-TIME SYSTEM (SSE)

Connection

```

GET /api/v1/sse/stream?token={jwt}
Response: text/event-stream

```

- Token via query param (EventSource API limitation — no custom headers)
- Keep-alive ping every 25 seconds (: ping\n\n)
- On connect: sends { type: 'connected', userId } confirmation

SSE Events

Event Name	Triggered By	Payload	Who Receives
location_update	POST /users/location	userId, latitude, longitude, accuracy, battery_level, timestamp	All members of user's circles
geofence_event	checkGeofenceStatus()	userId, userName, zoneName, eventType ('entry'/'exit'), timestamp	All circle members
sos_alert	POST /sos	userId, userName, userAvatar, latitude, longitude, message, timestamp	All circle members

admin_broadcast	POST /admin/broadcast	message, type ('info'/'warning'/'alert'), timestamp	ALL connected clients
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SSE Service (services/sse.js)

- `addClient(userId, res)` — registers SSE connection
- `removeClient(userId, res)` — removes on disconnect
- `sendToCircleMembers(circleId, event, data)` — queries DB for circle members → sends to each
- `sendToAllConnected(event, data)` — broadcasts to every connected client
- `getConnectedCount()` — returns total active SSE connections

Client-side SSE (ChildPanel)

```
const evtSource = new EventSource('/api/v1/sse/stream?token=' + token)
evtSource.addEventListener('location_update', handler)
evtSource.addEventListener('geofence_event', handler)
evtSource.addEventListener('sos_alert', handler)
evtSource.onerror = () => evtSource.close()
```

12. MEDIA / FILE UPLOAD SYSTEM

Avatar Upload Flow (3-step presigned upload)

```
Step 1: POST /api/v1/media/avatar/presign
Body: { contentType, fileSize }
Returns: { uploadUrl (R2 presigned), publicUrl }

Step 2: PUT {uploadUrl}
Headers: { Content-Type: file.type }
Body: file binary (no auth header – direct to R2)

Step 3: POST /api/v1/media/avatar/confirm
Body: { publicUrl }
Returns: { avatar_url }
→ Updates user.avatar_url in DB + localStorage
```

Storage: Cloudflare R2 (S3-compatible object storage) **Used in:** ChildPanel Profile tab, ParentPanel Settings

13. GEOFENCE SYSTEM

How it works

1. Parent creates a safe zone (circle on map) via ParentPanel
2. Stored in `safe_zones` table with PostGIS geometry
3. Every time a child posts location (`POST /users/location`), the backend calls `checkGeofenceStatus(userId, lat, lng)`
4. The service queries all safe zones for user's circles and checks if user is inside/outside each
5. State changes (inside → outside = "exit", outside → inside = "entry") fire:

- `geofence_event` SSE event to all circle members
- Log entry in `geofence_events` table

PostGIS Query (approximate)

```
SELECT sz.*, ST_Distance(
  sz.geom::geography,
  ST_SetSRID(ST_MakePoint($lng, $lat), 4326)::geography
) as distance_m
FROM safe_zones sz
JOIN circles c ON c.id = sz.circle_id
JOIN circle_members cm ON cm.circle_id = c.id
WHERE cm.user_id = $userId AND sz.active = TRUE
```

14. SOS ALERT SYSTEM

Trigger Sources

1. **ChildPanel Home tab** — hold SOS button 3 seconds
2. **ChildPanel SOS tab** — hold big SOS button 3 seconds
3. **ChildPanel SOS tab** — Quick Message buttons (instant send)
4. **(Future) Expo mobile app** — background SOS trigger

Flow

```
User holds SOS button (3s countdown)
  ↓
getCurrentPosition() → get fresh GPS
  ↓
POST /api/v1/sos { latitude, longitude, message }
  ↓
Backend:
  1. Gets all circles user belongs to
  2. SSE broadcasts 'sos_alert' to all circle members
  3. Logs to sos_events table
  4. Optionally logs GPS to device_locations
  ↓
Circle members receive SSE 'sos_alert':
- ChildPanel: toast + alert added to Alerts tab
- ParentPanel: alert + toast + can view location on map
- AdminPanel: visible in SOS Events tab
```

Cancel SOS

- Release button before 3 seconds = cancel (no API call)
- `cancelHomeSos()` / `cancelBigSos()` clears interval

15. ENVIRONMENT VARIABLES

```
# Backend (.env)
JWT_SECRET=<secret_key>           # JWT signing secret
ADMIN_SECRET=<admin_password>      # Admin panel password
DATABASE_URL=<postgres_url>        # PostgreSQL connection string

# SMS (optional – dev fallback to console.log)
MSG91_AUTH_KEY=<key>
MSG91_TEMPLATE_ID=<template>

# Cloudflare R2
R2_ACCOUNT_ID=<id>
R2_ACCESS_KEY_ID=<key>
R2_SECRET_ACCESS_KEY=<secret>
R2_BUCKET_NAME=<bucket>
R2_PUBLIC_URL=<cdn_url>

# Server
PORT=8002                          # Backend port
```

```
# Frontend (landing-react)
PORT=8090                          # Frontend server port
API_PORT=8002                      # Backend port to proxy to
```

16. SERVER SETUP

Frontend Server (landing-react/server.cjs)

- Plain Node.js HTTP server (no Express)
- Port: **8090**
- `/api/*` → proxy to backend:8002 (with SSE streaming support)
- Static files served from `dist/`
- SPA fallback: all unknown paths → `dist/index.html`
- MIME types handled manually

Backend Server (Express)

- Port: **8002**
- Rate limiting: 900s window / 1000 requests max
- Zod validation on all POST/PATCH routes
- PostGIS required (PostgreSQL extension)
- CORS handled by Express middleware

Build & Run

```
# Frontend
cd landing-react
npm install
npm run build      # Vite build → dist/
node server.cjs    # Start on :8090
```

```
# Backend
cd backend
npm install
node src/index.js    # Start on :8002
```

QUICK REFERENCE — WHAT EACH PANEL CAN DO

Feature	Child Panel	Parent Panel	Admin Panel
View own location	✓	—	—
Share GPS location	✓	✓	—
View family map	✓ (read-only)	✓ (full control)	—
Send SOS	✓	✓	—
Receive SOS	✓	✓	✓ (logs)
Create geofences	×	✓	—
View geofence alerts	✓	✓	✓ (logs)
Create family circle	×	✓	—
Join family circle	✓	—	—
Manage members	×	✓	✓
Upload avatar	✓	✓	—
View location history	✓ (own)	✓ (members)	—
Ban/unban users	×	×	✓
Delete users	×	×	✓
View OTP logs	×	×	✓
Broadcast messages	×	×	✓
Purge location data	×	×	✓
View system stats	×	×	✓

End of GRAVITY_SYSTEM_INFO.md